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• TEAM: SENSEISQUAD

Sensing & Power System — Component Justification Report

Comprehensive engineering rationale, trade-off analysis, power budget verification, and outdoor failure-mode mitigation for the AgriSpike field-node prototype.

Doc Version: 1.0.4

Architecture: ESP32 + SX1278 LoRa

Target Autonomy: 57 Days (Zero-Solar)



1. Purpose of This Report

This report documents the engineering rationale behind every sensor, power-chain component, and design decision in the current **AgriSpike** field-node prototype[cite: 7]. Its purpose is to demonstrate — with explicit reasoning, trade-off analysis, and calculated figures rather than assumed claims — that each component was deliberately selected, that the power system is feasible over realistic field-deployment periods, and that the design accounts for the failure modes inherent to low-cost sensors operating outdoors[cite: 7].

Explicit Measurement Standard:

Every figure presented as a *datasheet-typical estimate* is explicitly labelled as such[cite: 7]. Where this report states a figure derived from an actual measurement on the built prototype, it is explicitly labelled as *measured*[cite: 7]. This distinction is maintained throughout so that reviewers can verify exactly which claims are proven today and which are projected pending further field validation[cite: 7].



2. Sensing System — Component-by-Component Justification

Showing 10
Components

Evaluation of sensing units, voltage dividers, and communication modules[cite: 7].

COMPONENT	SYSTEM ROLE	ENGINEERING RATIONALE
DS18B20	Soil / near-surface temperature sensing	Digital 1-Wire output (single GPIO, multiple sensors on c

COMPONENT	SYSTEM ROLE	ENGINEERING RATIONALE
	(waterproof probe) [cite: 7].	bus); factory-calibrated; waterproof stainless probe suited to burial in soil[cite: 7].
DHT22 (AM2302)	Ambient air temperature and humidity[cite: 7].	Single package delivers two environmental readings needed for crop-stress and evapotranspiration context low cost and low pin count [7].
Resistive Soil Moisture v1.2	Volumetric soil moisture (relative) [cite: 7].	Lowest-cost option for zone level moisture sensing, compatible with ESP32 ADC input, easy to deploy at scale (zones)[cite: 7].
LDR (Photoresistor)	Relative ambient light level[cite: 7].	Cheap proxy for daylight/shade canopy-cover conditions correlated with evapotranspiration demand [7].
PIR Sensor	Human / animal intrusion detection near the node[cite: 7].	Adds a security/monitoring dimension (alerts on movement near valuable field hardware) low incremental cost[cite: 7].
Battery Divider (47kΩ + 47kΩ)	Battery voltage telemetry[cite: 7].	Lets each node self-report state-of-charge for remote diagnostics and predictive maintenance alerts[cite: 7].
Solar Divider (47kΩ + 47kΩ)	Solar panel output telemetry[cite: 7].	Confirms whether a node is actually receiving charge, distinguishing a shading/dust problem from a battery/ charging-circuit fault[cite: 7].

COMPONENT	SYSTEM ROLE	ENGINEERING RATIONALE
SX1278 (RA-02 433 MHz LoRa)	Long-range, low-power wireless uplink to the gateway[cite: 7].	Chosen over Wi-Fi/BLE for open-field km-scale line-of-sight range, and over 868/915 MHz for Indian ISM-band compliance and availability [7].
ESP32 (Node) / ESP32-S3 (Gateway)	Node Microcontroller / Gateway Microcontroller[cite: 7].	Standard ESP32 used at edge field nodes to minimize cost; higher-spec ESP32-S3 (with more RAM/PSRAM) reserved for central gateway[cite: 7].
Buzzer + Status LED	Local, no-laptop-required diagnostics[cite: 7].	Lets a technician confirm a node is alive/faulted directly in the field without requiring external diagnostics tools[cite: 7].



Integration Status Note:

The **PIR sensor** is included and active in the current node build[cite: 7].
The **NPK sensor** is not yet part of the current node — it is planned for the next-version prototype and will be demonstrated as a separate, standalone working module at review[cite: 7].



3. Power System Architecture & Rationale

Stage-by-stage engineering justification for power harvesting, charging, and regulation[cite: 7].

6V Solar Panel Selection

A 6V panel sits comfortably above the ~4.2–5V input range TP4056 chargers expect, leaving headroom to overcome wiring/diode losses without requiring an extra buck stage[cite: 7]. A 12V panel requires an added buck stage (extra cost and idle draw), while a 5V panel leaves insufficient headroom[cite: 7].

TP4056 Charge Module

Purpose-built, low-cost module consolidating single-cell CC/CV charging with on-board DW01A protection logic (over-charge, over-discharge, short-circuit protection) into one proven part[cite: 7].

2300mAh Li-ion Cell

Standard 18650-class cell[cite: 7]. Capacity was specifically calculated against measured/estimated draw to guarantee multi-week zero-solar autonomy during overcast weather[cite: 7].

3.3V Buck-Boost Converter

Maintains a stable 3.3V as Li-ion cell drops from 4.2V down to 3.0V[cite: 7]. Unlike buck-only regulators that drop out at ~3.5V (wasting ~30% capacity), buck-boost allows full capacity utilization and keeps ADC reference stable[cite: 7].



4. Power Budget — Feasibility Calculation

Calculated battery autonomy and solar recharge sufficiency[cite: 7].

4.1 Sleep-Mode Profile (~29 min 50s / cycle)

Element	Current (mA)	Basis
ESP32 Deep Sleep	0.01 – 0.15	Datasheet-typical[cite: 7]
Voltage Dividers (Gated)	~0 (vs ~70 µA ungated)	GPIO Transistor Gate[cite: 7]
DHT22 / DS18B20 Standby	~0.05	Datasheet-typical[cite: 7]
Total Sleep Draw	~0.06 – 0.20 mA	Sum (Gated design)[cite: 7]

4.2 Active-Mode Profile (~10s / cycle)

Element	Current (mA)	Basis
ESP32 Active + LoRa TX	~240	Worst-case (+20dBm)[cite: 7]
Active Sensors Read	~20 – 25	Combined sensors[cite: 7]
Total Active Draw	~260 – 265 mA	10-second wake window[cite: 7]

4.3 & 4.4 Autonomy Derivation

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// Daily Active Contribution (48 cycles/day)

Active = 260 mA × (10s / 3600s) × 48 = 0.72 mAh/day[cite: 7]

// Daily Sleep Contribution (24 hrs continuous)

Sleep = 0.15 mA × 24 hrs ≈ 3.60 mAh/day[cite: 7]
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Total Daily Draw ≈ 40 mAh/day per node[cite: 7]

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// Battery Autonomy (Zero Solar Input):

2300 mAh ÷ 40 mAh/day = 57 Days Autonomy[cite: 7]
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This 57-day buffer guarantees that prolonged monsoon cloud cover will not cause node browns[cite: 7]. The 3.3V buck-boost converter makes this entire 57-day figure usable by preventing dropout above 3.0V[cite: 7].

☀️ 4.5 Solar Recharge Sufficiency Calculation (6V, 100mA Panel)

Step 1: Raw Harvest

$100\text{mA} \times 4.5 \text{ Peak Sun Hrs} = \mathbf{450 \text{ mAh/day}}$ [cite: 7]

Step 2: Real-World Derate

$450 \text{ mAh} \times 65\% \text{ (Dust/Heat)} = \mathbf{293 \text{ mAh/day}}$ [cite: 7]

Step 3: Charger Efficiency

$293 \text{ mAh} \times 80\% \text{ (TP4056 loss)} = \mathbf{234 \text{ mAh/day}}$ [cite: 7]

Step 4: Energy Margin

$234 \text{ mAh vs } 40 \text{ mAh} = \mathbf{5.8\times \text{ Net Surplus}}$ [cite: 7]

Even during heavy monsoons reducing solar yield to 20% (~47 mAh/day), the solar harvest alone still covers daily node consumption, maintaining complete sustainability[cite: 7].



5. Recommendations for Strengthening Report Before Submission

REC-01: Multimeter Verification

Replace estimated current figures with physical multimeter/power profiler measurements taken directly on the prototype to upgrade claims from 'estimated' to 'measured'[cite: 7].

REC-02: Solar Panel Rating Label

Supply the exact panel rating label (wattage/current) to finalize Section 4.5 using exact hardware specifications[cite: 7].

REC-03: NPK Module Demo Alignment

Ensure live evaluation explicitly demonstrates the NPK sensor as a standalone working module so the demo matches document claims[cite: 7].

REC-04: Confidence Matrix

Add a 'confidence level' note next to each pro/con entry to distinguish directly observed field behaviors from datasheet specifications[cite: 7].